VLSI Devices
Lecture 9

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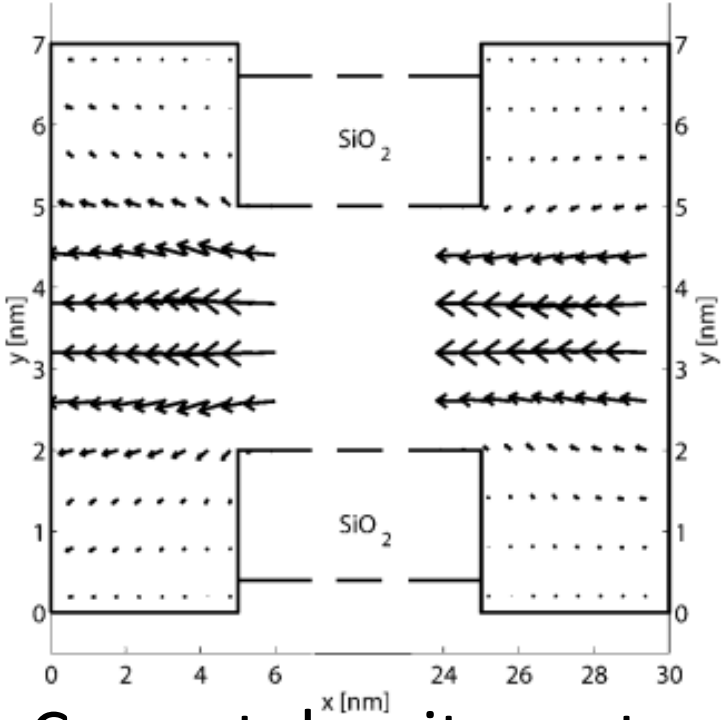
L9

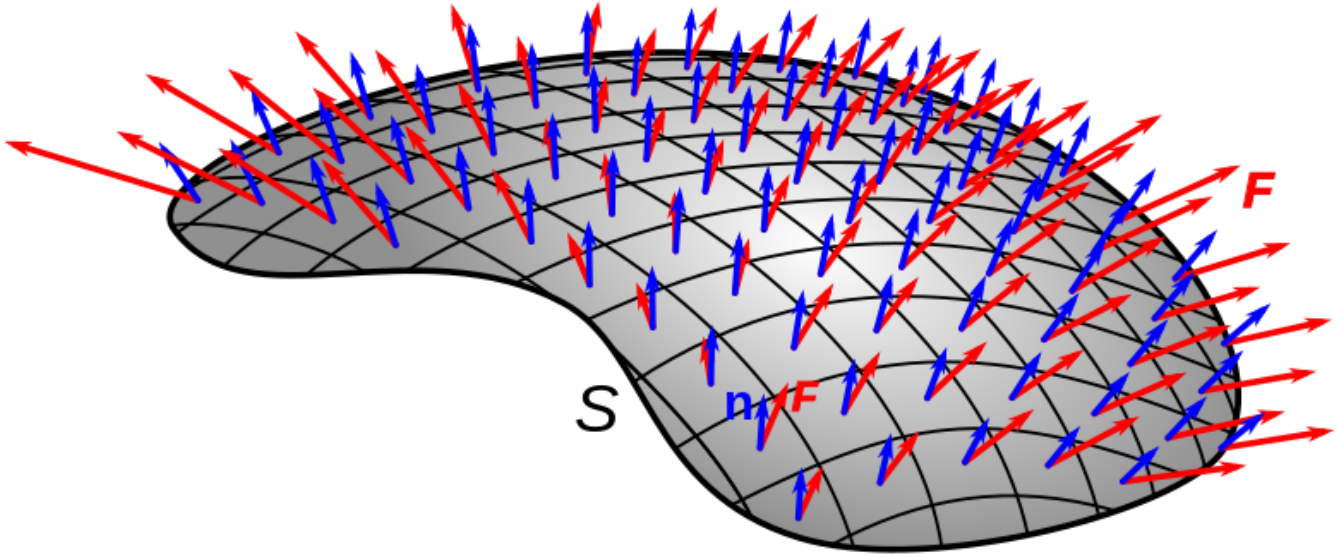
Terminal current vs. current density

- $\mathbf{J}$, current per area (defined everywhere)

$$I_{terminal} = -\int_{terminal} \mathbf{J} \cdot d\mathbf{a}$$



Current density vector (Luisier et al., JAP, vol. 100, p. 043713, 2006)



Surface integral of a vector field (Wikipedia)

Resistivity & sheet resistivity

- Drift current density

$$\mathbf{J}_{n,drift} = q\mu_n n\mathbf{E}$$ Taur, Eq. (2.34)

$$\mathbf{J}_{p,drift} = q\mu_p p\mathbf{E}$$ Taur, Eq. (2.36)

  - Then, the total resistivity is

Ω cm

$$\rho = \frac{1}{q\mu_n n + q\mu_p p}$$ Taur, Eq. (2.38)

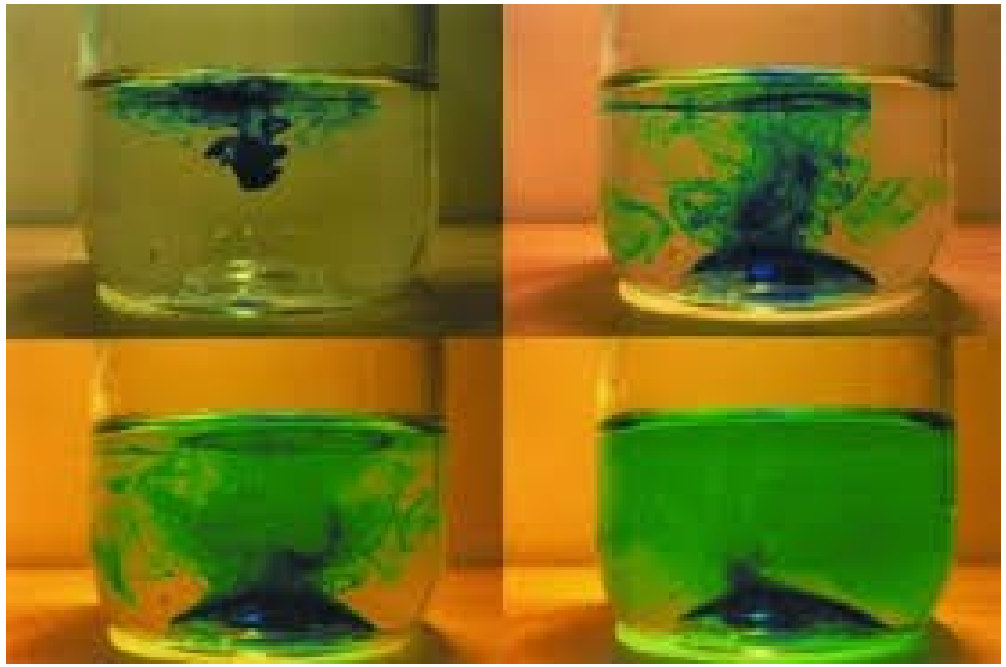
- Sheet resistivity
  - The resistance of a conductor of length, L, and width, W,

$$R = \frac{L}{W}\rho_{sh}$$ Taur, Eq. (2.41)

Ω/□

Diffusion

- It is not only for charged particles.
  - For example,



Diffusion of ink
(Google images)

  - Therefore, no polarity is expected.

Equation

- Flux
  - The electron flux due to the diffusion mechanism is given by

$$\mathbf{F}_n = -D_n \nabla n$$

  where D_n is the electron diffusion coefficient in the unit of (cm^2/sec).
  - The diffusion current density is

$$\mathbf{J}_{n,diff} = qD_n \nabla n$$

Taur, Eq. (2.46)

- How about the hole?
  - The diffusion current density is

$$\mathbf{J}_{p,diff} = -qD_p \nabla p$$

Taur, Eq. (2.47)

Expressions for current densities

- Total current density
  - Electron current density

$$\mathbf{J}_n = q\mu_n n\mathbf{E} + qD_n\nabla n$$

Taur, Eq. (2.68)

  - Hole current density

$$\mathbf{J}_p = q\mu_p p\mathbf{E} - qD_p\nabla p$$

Taur, Eq. (2.69)

  - Displacement current density

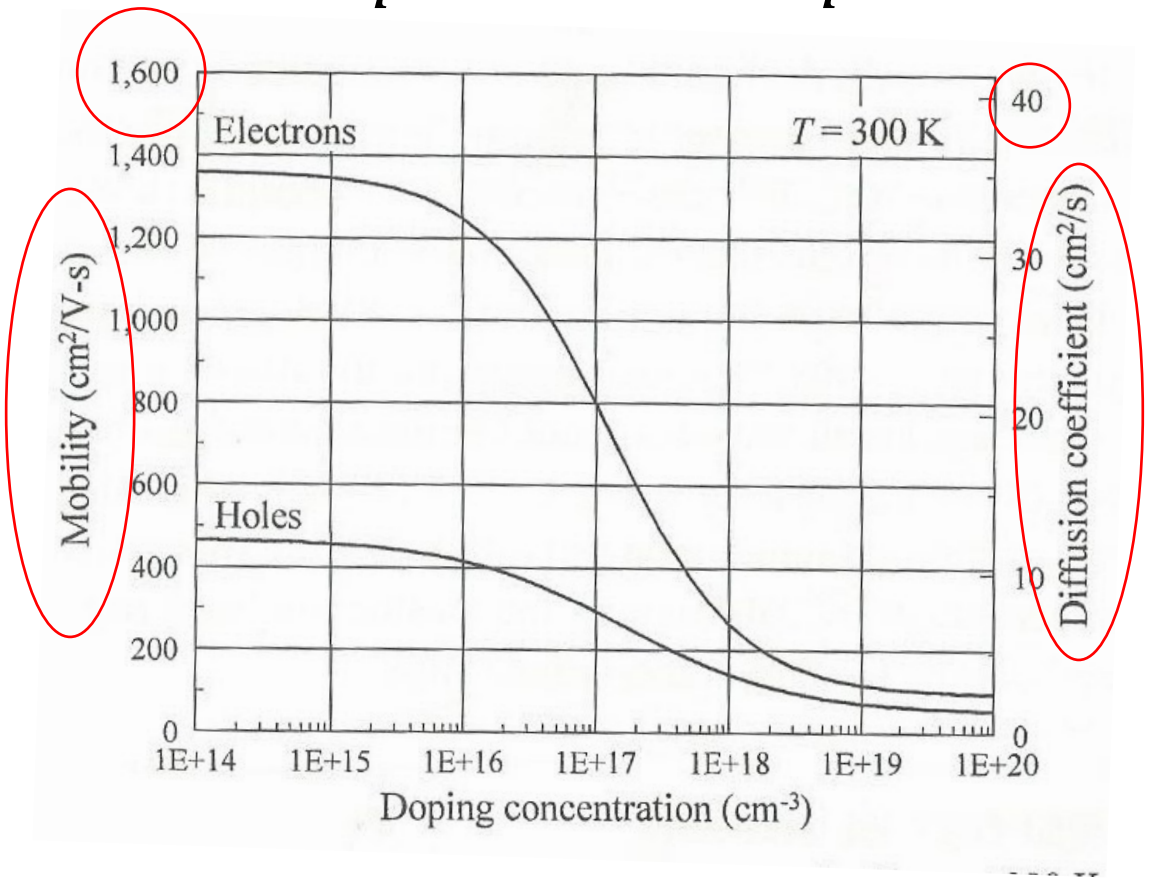
$$\mathbf{J}_{displacement} = \frac{\partial}{\partial t}(\epsilon\mathbf{E})$$

Einstein relation

- At equilibrium, we have the following relations:

$$D_n = \frac{k_B T}{q}\mu_n, D_p = \frac{k_B T}{q}\mu_p$$

Taur, Eq. (2.52) and Eq. (2.53)



(Y. J. Park's book)

Current density with Einstein relation

- Alternative forms
  - Electron current density

$$\mathbf{J}_n = -q\mu_n n\left[\nabla\phi - \frac{k_B T}{q}\frac{1}{n}\nabla n\right]$$

Taur, Eq. (2.70)

  - Hole current density

$$\mathbf{J}_p = -q\mu_p p\left[\nabla\phi + \frac{k_B T}{q}\frac{1}{p}\nabla p\right]$$

Taur, Eq. (2.71)

Revisiting carrier concentrations

$E_f = -q\phi_f$

- Carrier densities are expressed as

$$n = n_i \exp\left(-\frac{E_i - E_f}{k_B T}\right) = n_i \exp\left(\frac{\phi - \phi_f}{k_B T/q}\right)$$ Taur, Eq. (2.63)

$$p = n_i \exp\left(\frac{E_i - E_f}{k_B T}\right) = n_i \exp\left(-\frac{\phi - \phi_f}{k_B T/q}\right)$$ Taur, Eq. (2.64)

  – These relations are generally applicable in the presence of net charge and band bending.

Extension to non-equilibrium cases

- Of course, at non-equilibrium cases, we cannot define E_f.
  - However, we *introduce* ϕ_n and ϕ_p to satisfy:

$$n = n_i \exp\left(\frac{\phi - \phi_n}{k_B T/q}\right)$$ Taur, Eq. (2.75)

$$p = n_i \exp\left(-\frac{\phi - \phi_p}{k_B T/q}\right)$$ Taur, Eq. (2.76)

  - They are called quasi-Fermi potentials. (Electron quasi-Fermi potential and hole quasi-Fermi potential)

Current density with $\nabla\phi_n$ and $\nabla\phi_p$

- From the expression for the quasi-Fermi potential,
  - It can be written as

$$\phi_n = \phi - \frac{k_B T}{q} \log \frac{n}{n_i}$$

Taur, Eq. (2.79)

  - Taking the gradient,

$$\nabla\phi_n = \nabla\phi - \frac{k_B T}{q} \frac{\nabla n}{n}$$

  - Using this expression,

$$\mathbf{J}_n = -q\mu_n n \left[\nabla\phi - \frac{k_B T}{q} \frac{1}{n} \nabla n \right] = -q\mu_n n \nabla\phi_n$$

Taur, Eq. (2.77)

$$\mathbf{J}_p = -q\mu_p p \nabla\phi_p$$

Taur, Eq. (2.78)

Gradient of quasi-Fermi potential

- We have the following relations,

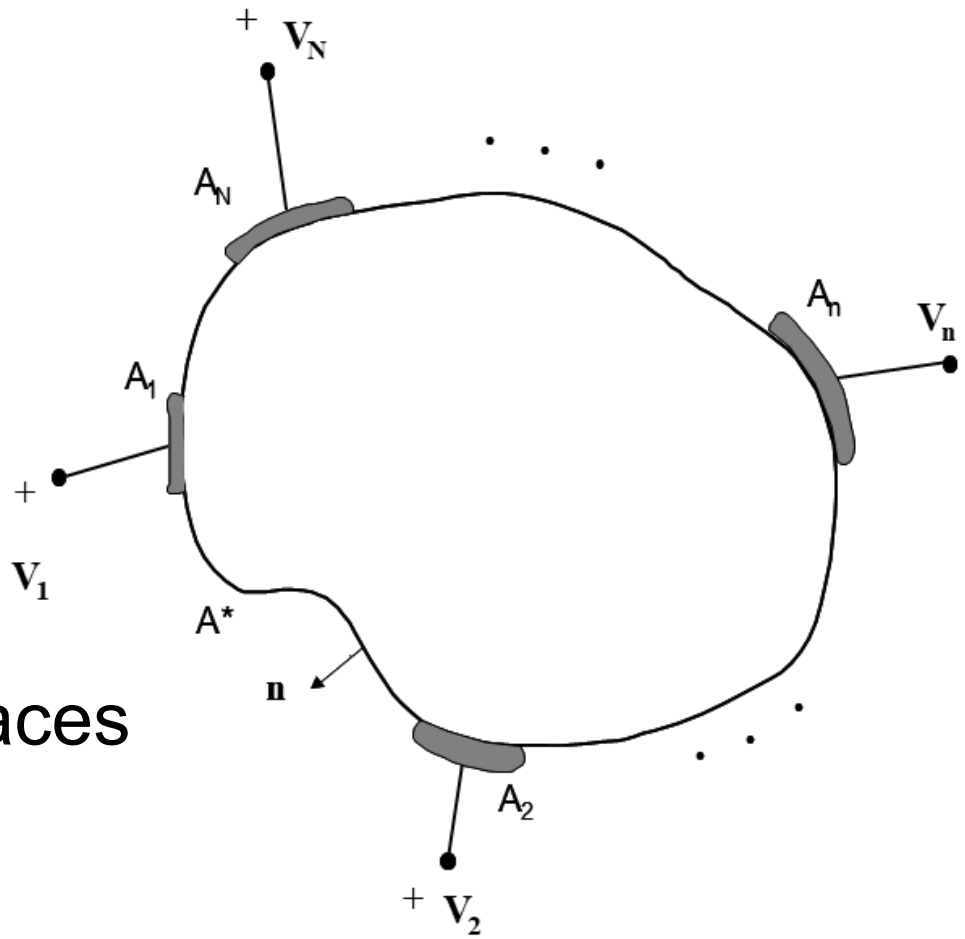
$$\mathbf{J}_n = -\, q\mu_n n \nabla \phi_n$$

$$\mathbf{J}_p = -q\mu_p p \nabla \phi_p$$

  - The gradient of electron quasi-Fermi potential drives the electron current.
  - The gradient of hole quasi-Fermi potential drives the hole current.

Boundary conditions for ϕ_n and ϕ_p

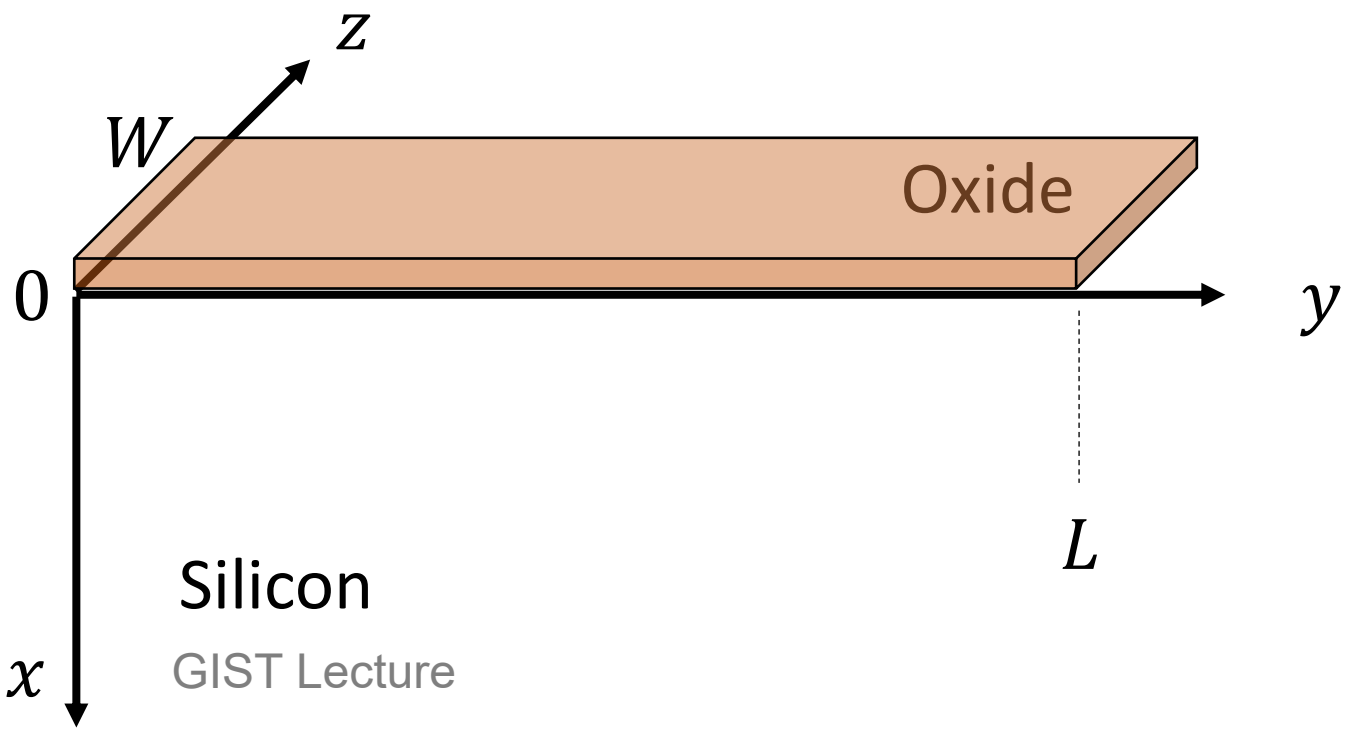
- Dirichlet boundary condition
  - At terminals,
    $\phi_n = \phi_p = V_{app}$



- Neumann boundary condition
  - No current through non-contact surfaces
    $\nabla\phi_n \cdot \mathbf{n} = \nabla\phi_p \cdot \mathbf{n} = 0$

Schematic of channel region (Planar)

- $x = 0$ at silicon surface
  - $y = 0$ at the source and $y = L$ at the drain
  - Source and substrate are grounded.
  - Uniform p-type substrate



Gradual channel approximation (GCA)

- Variation of the electric field in the y-direction is much less than the corresponding variation in the x-direction.

$$\left|\frac{\partial^2\phi}{\partial x^2}\right| \gg \left|\frac{\partial^2\phi}{\partial y^2}\right|$$

  - Poisson equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = -\frac{q}{\epsilon_{si}}[p(x,y) - n(x,y) - N_a]$$

  - Poisson equation under the GCA

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\epsilon_{si}}[p(x,y) - n(x,y) - N_a]$$

Electron quasi-Fermi potential, $V(y)$

- It is assumed that V is independent of x.
  - *Why?* MOSFET current flows predominantly along the y-direction.
  - Since $\mathbf{J}_n = -q\mu_n n \nabla \phi_n$, V varies mainly along the y-direction.
  - Boundary conditions:

$$V(y=0) = V_s = 0$$
$$V(y=L) = V_d = V_{ds}$$

- Electron density, $n(x,y)$

$$n(x,y) = \frac{n_i^2}{N_a} \exp\left(\frac{q}{k_B T}(\phi - V)\right)$$

Taur, Eq. (5.1)

Still, ϕ_B is the reference value.

Our previous expressions (1)

- They are modified by V. (Only terms related with the electron density)
  - Electric field

$$E_x^2(x,y) = \left(\frac{d\phi}{dx}\right)^2$$
$$= \frac{2k_BTN_a}{\epsilon_{si}}\left[\left(\exp\left(-\frac{q\phi}{k_BT}\right)+\frac{q\phi}{k_BT}-1\right)\right.$$
$$\left.+\frac{n_i^2}{N_a^2}\left(\exp\left(-\frac{qV}{k_BT}\right)\left(\exp\left(\frac{q\phi}{k_BT}\right)-1\right)-\frac{q\phi}{k_BT}\right)\right]$$

Taur, Eq. (5.2)

Our previous expressions (2)

- They are modified by V. (Only terms related with the electron density)
  - Surface inversion

$$\phi(0, y) = V(y) + 2\phi_B$$

Taur, Eq. (5.3)

  - Maximum depletion layer width

$$W_{dm} = \sqrt{\frac{2\epsilon_{si}[V(y) + 2\phi_B]}{qN_a}}$$

Taur, Eq. (5.4)

- Summary: With the GCA, our MOS expressions are re-used only with modification by V.

Drain current

$$\mathbf{J}_n = -q\mu_n n \nabla \phi_n$$

- Electron current density at a point (x, y)

$$J_n(x,y) = -q\mu_n n(x,y)\frac{dV}{dy}$$ Taur, Eq. (5.5)

  - (It includes both the drift and diffusion currents.)
  - When integrated from $x = 0$ to x_i, (and from $z = 0$ to W)

$$I_d(y) = qW\int_0^{x_i} \mu_n n(x,y)\frac{dV}{dy}dx$$ Taur, Eq. (5.6)

Sign change due to convention of terminal current

z-directional width

Further simplification

- Electron current density at a point (x, y)

$$I_d(y) = qW \int_0^{x_i} \mu_n n(x,y) \frac{dV}{dy} dx = -\mu_{eff} W \frac{dV}{dy} \left(-q \int_0^{x_i} n(x,y) dx \right)$$

$$= -\mu_{eff} W \frac{dV}{dy} Q_i(y)$$

Taur, Eq. (5.8)

  - We introduce an effective mobility, μ_{eff}.
  - Since V is a function of y only, V is interchangeable with y.

$$Q_i(y) = Q_i(V)$$

  - Then,

$$I_d(y) dy = \mu_{eff} W [-Q_i(V)] dV$$

$I_d(y)$ is actually a constant.

- When integrated from $y = 0$ to L, (from $V = 0$ to V_{ds})

$$I_d = \mu_{eff} \frac{W}{L} \int_0^{V_{ds}} [-Q_i(V)] dV$$

Taur, Eq. (5.10)

  - Then, how can we find $Q_i(V)$? (We must perform the x-directional integration.)

$$Q_i = -q \int_0^{x_i} n(x, y) dx = -q \int_{\phi_s}^{\delta} n(\phi, V) \frac{dx}{d\phi} d\phi$$

It's small, but not zero.

$$= -q \int_{\delta}^{\phi_s} \frac{\left(n_i^2 / N_a\right) \exp\left(\frac{q}{k_B T}(\phi - V)\right)}{E_x(\phi, V)} d\phi$$

Taur, Eq. (5.12)

Pao-Sah double integral

- Finally, the expression for I_d reads

$$I_d = q\mu_{eff}\frac{W}{L}\int_0^{V_{ds}}\left[\int_\delta^{\phi_s}\frac{\left(n_i^2/N_a\right)\exp\left(\frac{q}{k_BT}(\phi - V)\right)}{E_x(\phi, V)}d\phi\right]dV$$

Taur, Eq. (5.13)

  - It is the Pao-Sah double integral.
  - Rigorous within the GCA, but it is difficult to evaluate.

How can we determine ϕ_s?

- For given V_{gs} and V, we can solve the MOS equation.

$$V_{gs} = V_{fb} + \phi_s - \frac{Q_s}{C_{ox}}$$

$$= V_{fb} + \phi_s + \frac{\sqrt{2\epsilon_{si} k_B T N_a}}{C_{ox}} \left[\frac{q\phi_s}{k_B T} + \frac{n_i^2}{N_a^2} \exp\left(\frac{q}{k_B T} (\phi_s - V) \right) \right]^{1/2}$$

Taur, Eq. (5.14)

Only two important terms are kept.

  - We can numerically solve the above equation to obtain ϕ_s. (Newton method)

Thank you!